

K-Nearest Neighbors (KNN) and Cosine Similarity

Step-by-Step Explanation with Calculations

1 K-Nearest Neighbors (KNN)

K-Nearest Neighbors (KNN) is a **supervised machine learning algorithm** used for:

- Classification
- Regression

It is called a **lazy learning algorithm** because it does not build a model during training. Instead, it stores all training data and performs computation only at prediction time.

1.1 Working Steps of KNN

1. Choose the number of neighbors K
2. Calculate distance or similarity between test point and all training points
3. Select the K nearest neighbors
4. Assign the class by majority voting

2 Cosine Similarity

Cosine similarity measures how similar two vectors are by calculating the **cosine of the angle** between them.

2.1 Formula

$$\text{Cosine Similarity}(A, B) = \frac{A \cdot B}{\|A\| \|B\|}$$

3 Solved Numerical Example (Cosine Similarity)

3.1 Given Vectors

$$A = (1, 2, 3), \quad B = (2, 1, 2)$$

3.2 Step 1: Dot Product Calculation

$$A \cdot B = (1 \times 2) + (2 \times 1) + (3 \times 2)$$

$$A \cdot B = 2 + 2 + 6 = 10$$

3.3 Step 2: Magnitude of Vector A

$$\|A\| = \sqrt{1^2 + 2^2 + 3^2}$$

$$\|A\| = \sqrt{1 + 4 + 9}$$

$$\|A\| = \sqrt{14}$$

3.4 Step 3: Magnitude of Vector B

$$\|B\| = \sqrt{2^2 + 1^2 + 2^2}$$

$$\|B\| = \sqrt{4 + 1 + 4}$$

$$\|B\| = \sqrt{9} = 3$$

3.5 Step 4: Cosine Similarity

$$\cos(A, B) = \frac{10}{\sqrt{14} \times 3}$$

$$\cos(A, B) = \frac{10}{11.22} \approx 0.89$$

Cosine Similarity = 0.89

4 KNN Classification Using Cosine Similarity

4.1 Training Data

Point	Vector	Class
P1	(1,0,1)	A
P2	(0,1,1)	B
P3	(1,1,0)	A

4.2 Test Vector

$$T = (1, 1, 1)$$

4.3 Cosine Similarity with $P1$

$$T \cdot P1 = (1 \times 1) + (1 \times 0) + (1 \times 1) = 2$$

$$\|T\| = \sqrt{1^2 + 1^2 + 1^2} = \sqrt{3}$$

$$\|P1\| = \sqrt{1^2 + 0^2 + 1^2} = \sqrt{2}$$

$$\cos(T, P1) = \frac{2}{\sqrt{3}\sqrt{2}} = \frac{2}{2.45} \approx 0.816$$

4.4 Cosine Similarity with $P2$

$$T \cdot P2 = (1 \times 0) + (1 \times 1) + (1 \times 1) = 2$$

$$\|P2\| = \sqrt{0^2 + 1^2 + 1^2} = \sqrt{2}$$

$$\cos(T, P2) = \frac{2}{\sqrt{3}\sqrt{2}} \approx 0.816$$

4.5 Cosine Similarity with $P3$

$$T \cdot P3 = (1 \times 1) + (1 \times 1) + (1 \times 0) = 2$$

$$\|P3\| = \sqrt{1^2 + 1^2 + 0^2} = \sqrt{2}$$

$$\cos(T, P3) = \frac{2}{\sqrt{3}\sqrt{2}} \approx 0.816$$

4.6 Final Decision

For $K = 3$:

$$\text{Classes} = \{A, B, A\}$$

$$\boxed{\text{Predicted Class} = A}$$

5 Real World Example: Text Classification

Vocabulary:

$$\{\text{cat}, \text{dog}, \text{milk}, \text{pet}\}$$

5.1 Encoded Sentences

Sentence	Vector	Class
cat drinks milk	(1,0,1,0)	Animal
dog is pet	(0,1,0,1)	Animal
milk product	(0,0,1,0)	Food

Test sentence: **cat pet**

$$T = (1, 0, 0, 1)$$

5.2 Cosine Similarity with Sentence 1

$$T \cdot S1 = (1 \times 1) + (0 \times 0) + (0 \times 1) + (1 \times 0) = 1$$

$$\|T\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\|S1\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\cos(T, S1) = \frac{1}{2} = 0.5$$

5.3 Cosine Similarity with Sentence 2

$$T \cdot S2 = (1 \times 0) + (0 \times 1) + (0 \times 0) + (1 \times 1) = 1$$

$$\cos(T, S2) = \frac{1}{2} = 0.5$$

5.4 Final Classification

For $K = 2$:

Predicted Class = Animal

6 Curse of Dimensionality

As dimensions increase:

- Distances become meaningless
- Data becomes sparse
- KNN accuracy decreases

Cosine similarity helps because it depends on direction, not magnitude.

7 Document Representation and Cosine Similarity (Detailed Example)

7.1 Vocabulary

Consider a small vocabulary:

$$\text{Vocab} = \{\text{cat}, \text{dog}\}$$

Each document is represented using a **term-frequency (TF) vector** of the form:

$$(\text{count}(\text{cat}), \text{count}(\text{dog}))$$

7.2 Document Vectors

- **Doc1:** “cat cat dog”
 $\Rightarrow (2, 1)$
- **Doc2:** “cat dog”
 $\Rightarrow (1, 1)$
- **Doc3:** “dog dog dog dog”
 $\Rightarrow (0, 4)$

7.3 Cosine Similarity Formula

Cosine similarity measures the angle between two vectors:

$$\cos(\theta) = \frac{A \cdot B}{\|A\| \|B\|}$$

where

$$A \cdot B = \sum A_i B_i, \quad \|A\| = \sqrt{\sum A_i^2}$$

7.4 Similarity Between Doc1 and Doc2

$$\text{Doc1} = (2, 1), \quad \text{Doc2} = (1, 1)$$

Step 1: Dot Product

$$(2 \times 1) + (1 \times 1) = 2 + 1 = 3$$

Step 2: Magnitude of Doc1

$$\|\text{Doc1}\| = \sqrt{2^2 + 1^2} = \sqrt{4 + 1} = \sqrt{5}$$

Step 3: Magnitude of Doc2

$$\|\text{Doc2}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

Step 4: Cosine Similarity

$$\cos(\text{Doc1}, \text{Doc2}) = \frac{3}{\sqrt{5}\sqrt{2}} = \frac{3}{\sqrt{10}}$$

$$\cos(\text{Doc1}, \text{Doc2}) \approx 0.95$$

7.5 Similarity Between Doc1 and Doc3

$$\text{Doc1} = (2, 1), \quad \text{Doc3} = (0, 4)$$

Step 1: Dot Product

$$(2 \times 0) + (1 \times 4) = 4$$

Step 2: Magnitude of Doc1

$$\|\text{Doc1}\| = \sqrt{5}$$

Step 3: Magnitude of Doc3

$$\|\text{Doc3}\| = \sqrt{0^2 + 4^2} = \sqrt{16} = 4$$

Step 4: Cosine Similarity

$$\cos(\text{Doc1}, \text{Doc3}) = \frac{4}{\sqrt{5} \times 4} = \frac{4}{\sqrt{80}}$$

$$\cos(\text{Doc1}, \text{Doc3}) \approx 0.45$$

7.6 Effect of Document Length (Normalization)

Consider a longer version of Doc2:

$$\text{Doc2}_{\text{long}} = (10, 10)$$

Step 1: Dot Product

$$(1 \times 10) + (1 \times 10) = 20$$

Step 2: Magnitudes

$$\|\text{Doc2}\| = \sqrt{1^2 + 1^2} = \sqrt{2}$$

$$\|\text{Doc2}_{\text{long}}\| = \sqrt{10^2 + 10^2} = \sqrt{200}$$

Step 3: Cosine Similarity

$$\cos(\text{Doc2}, \text{Doc2}_{\text{long}}) = \frac{20}{\sqrt{2} \times \sqrt{200}}$$

$$\cos(\text{Doc2}, \text{Doc2}_{\text{long}}) = \frac{20}{1.41 \times 14.14} = \frac{20}{20} = 1$$

7.7 Final Interpretation

- Cosine similarity is **independent of document length**
- Doc1 is more similar to Doc2 (0.95) than to Doc3 (0.45)
- Identical documents always give similarity = 1

Cosine similarity captures semantic similarity effectively

8 Error vs K in KNN (Numerical Example)

We consider a simple **binary classification** problem to understand how the classification error of KNN changes with different values of K .

8.1 Dataset

Point	Feature Value (x)	True Class
P1	1	+
P2	2	+
P3	3	+
P4	6	-
P5	7	-
P6	8	-

Test point:

$$x_{\text{test}} = 4$$

True class of test point:

$$y_{\text{true}} = +$$

Distance metric used:

$$d(x_i, x_{\text{test}}) = |x_i - x_{\text{test}}|$$

8.2 Distance Computation

Training Point	Distance from x_{test}	Class
P1 (1)	$ 1 - 4 = 3$	+
P2 (2)	$ 2 - 4 = 2$	+
P3 (3)	$ 3 - 4 = 1$	+
P4 (6)	$ 6 - 4 = 2$	-
P5 (7)	$ 7 - 4 = 3$	-
P6 (8)	$ 8 - 4 = 4$	-

Sorted neighbors by distance:

$$(+, +, -, +, -, -)$$

8.3 Prediction and Error for Different Values of K

8.3.1 $K = 1$

Nearest neighbor: (+)

Prediction:

$$\hat{y} = +$$

Error:

$$E = 0$$

8.3.2 $K = 3$

Nearest neighbors:

$(+, +, -)$

Majority class: $(+)$

$$E = 0$$

8.3.3 $K = 5$

Nearest neighbors:

$(+, +, -, +, -)$

Majority class: $(+)$

$$E = 0$$

8.3.4 $K = 6$

All neighbors considered:

Votes:

$$+ = 3, \quad - = 3$$

Assuming tie breaks toward class $(-)$:

Prediction:

$$\hat{y} = -$$

Error:

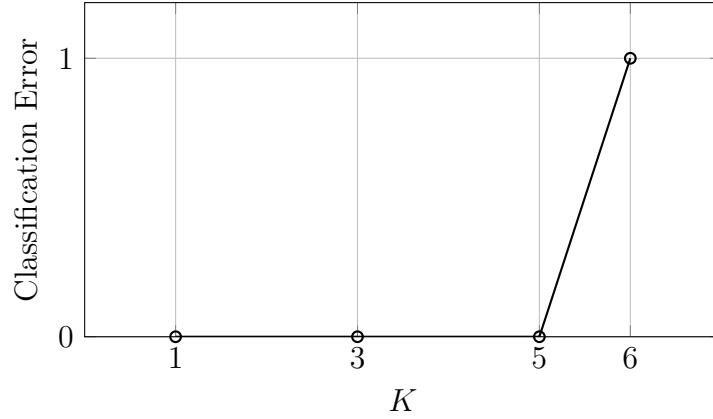
$$E = 1$$

8.4 Error Table

K	Predicted Class	Error
1	+	0
3	+	0
5	+	0
6	-	1

8.5 Graph: Error vs K

Note: The graph below shows how error changes with increasing K .



8.6 Observation

- Small K gives low bias but high variance
- Moderate K gives minimum classification error
- Large K increases bias and causes underfitting

8.7 Conclusion

As the value of K increases, the classification error of KNN first decreases and then increases, forming a **U-shaped error curve**. This behavior illustrates the **bias–variance tradeoff** in the KNN algorithm.

9 Numerical Example: Selection of Optimal k in k-NN and Bayes Error

Consider a dataset with 6 data points, each labeled with one of two classes: A or B. The feature values and true labels are given in Table 1.

Data Point	Feature (x)	Class
x_1	2	A
x_2	4	A
x_3	5	B
x_4	7	B
x_5	8	B
x_6	9	A

Table 1: Dataset for k-NN example

9.1 k-NN Predictions and Error Rates

The goal is to select the optimal value of k using the following procedure:

1. For each value of k , perform k-NN prediction for every data point.

2. Assign the class label by majority vote among the k nearest neighbors.
3. Compute the misclassification rate (error) as:

$$\text{Error}(k) = \frac{\text{Number of misclassified points}}{6}.$$

4. Compare errors for all tested values of k and choose the one with the lowest misclassification rate as optimal.

Step-by-Step Computation

• For $k = 1$:

- x_1 's nearest neighbor: x_2 (A) $\Rightarrow$ Correct
- x_2 's nearest neighbor: x_1 (A) $\Rightarrow$ Correct
- x_3 's nearest neighbor: x_2 (A) $\Rightarrow$ Misclassified
- x_4 's nearest neighbor: x_3 (B) $\Rightarrow$ Correct
- x_5 's nearest neighbor: x_4 (B) $\Rightarrow$ Correct
- x_6 's nearest neighbor: x_5 (B) $\Rightarrow$ Misclassified

$$\text{Error}(k = 1) = \frac{2}{6} \approx 0.33$$

• For $k = 2$:

- x_1 : x_2 (A), x_3 (B) $\Rightarrow$ Majority: A $\Rightarrow$ Correct
- x_2 : x_1 (A), x_3 (B) $\Rightarrow$ Tie $\Rightarrow$ pick x_1 $\Rightarrow$ Correct
- x_3 : x_2 (A), x_4 (B) $\Rightarrow$ Tie $\Rightarrow$ pick x_4 $\Rightarrow$ Correct
- x_4 : x_3 (B), x_5 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Correct
- x_5 : x_4 (B), x_6 (A) $\Rightarrow$ Tie $\Rightarrow$ pick x_4 $\Rightarrow$ Correct
- x_6 : x_5 (B), x_4 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Misclassified

$$\text{Error}(k = 2) = \frac{1}{6} \approx 0.17$$

• For $k = 3$:

- x_1 : x_2 (A), x_3 (B), x_4 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Misclassified
- x_2 : x_1 (A), x_3 (B), x_4 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Misclassified
- x_3 : x_2 (A), x_4 (B), x_5 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Correct
- x_4 : x_3 (B), x_5 (B), x_6 (A) $\Rightarrow$ Majority: B $\Rightarrow$ Correct
- x_5 : x_4 (B), x_6 (A), x_3 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Correct
- x_6 : x_5 (B), x_4 (B), x_3 (B) $\Rightarrow$ Majority: B $\Rightarrow$ Misclassified

$$\text{Error}(k = 3) = \frac{2}{6} \approx 0.33$$

k	Error Rate
1	0.33
2	0.17
3	0.33

Table 2: Error rates for different k values

Error Rates Summary

The optimal k is $k = 2$, as it achieves the lowest misclassification rate.

9.2 Bayes Error Calculation

The **Bayes error** represents the minimum possible error achievable by any classifier, given the underlying class distributions. Assuming the class distributions are as follows (for simplicity in this 1D example):

- $P(A) = 3/6 = 0.5$, $P(B) = 3/6 = 0.5$
- We assume Gaussian-like overlapping regions around the feature values.

A simple approximation of Bayes error in 1D discrete space:

$$\text{Bayes Error} = \sum_{x \in \text{overlap region}} \min\{P(A|x), P(B|x)\}$$

Looking at our data, the overlap occurs around $x = 5$ (between A and B):

$$\text{Bayes Error} \approx \frac{1}{6} \approx 0.17$$

This matches the error achieved by $k = 2$, confirming that k-NN with $k = 2$ is near-optimal for this dataset.

Conclusion

KNN with cosine similarity is highly effective for high-dimensional data such as text, where traditional distance measures fail.